



PAK-AUSTRIA FACHHOCHSCHULE

INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY



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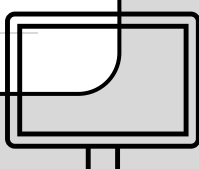
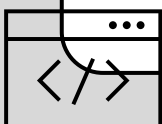
Program : Cybersecurity

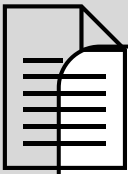
Instructor's name: Hassan Ali

Date Submitted : 13 / 07 / 26

Deadline : 13 / 07 / 26

Assignment Report # 6





• INTRODUCTION •

In this task we learned about a few efficient commands for managing your operating system. The main objective was to run a few commands on the terminal. We were given an assignment at the end; it goes as follows.

Assignments Tasks:

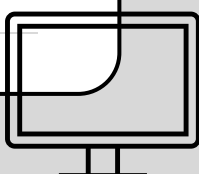
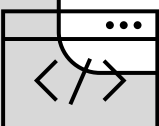
Task : *Run Commands*, Update the OS. Check disk usage. Check memory usage. Manage a service. Monitor running processes.

• TASK •

Task : *Run Commands*, Update the OS. Check disk usage. Check memory usage. Manage a service. Monitor running processes.

Steps Followed:

- Run command for Updating the OS.
- Run command for Checking Disk Usage.
- Run command for Checking Memory Usage.
- Run command for Manage a Service.
- Run command for Monitor Running Processes.



• STEP # 01 •

Step # 01 : Run command for Updating the OS.

- Open terminal in kali linux.
- Type the command ***sudo apt update && sudo apt upgrade -y***.
- Output is as follows.

```
(kali@kali)-[/home/kali]
PS> sudo apt update && sudo apt upgrade -y
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
All packages are up to date.
The following packages were automatically installed and are no longer require
d:
  firmware-marvell-prestera libopenjph0.27 libvidstab1.1 libx265-215
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

Step #01: Command `sudo apt update && sudo apt upgrade -y`

This will first check for any packages then update each one.

• STEP # 02 •

Step # 02 : Run command for Checking Disk Usage.

- Open terminal in kali linux.
- Type the command ***df -h***.
- Output is as follows.

```
(kali@kali)-[/home/kali]
PS> df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            871M   0    871M   0% /dev
tmpfs           194M  1.2M   193M   1% /run
/dev/sda1       79G   17G   58G   23% /
tmpfs           967M  4.0K   967M   1% /dev/shm
tmpfs           967M  3.4M   964M   1% /tmp
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  108K   194M   1% /run/user/1000
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.servi
ce
```

Step #02: Command `df -h`

This will display disk usage.

• STEP # 03 •

Step # 03 : Run command for Checking Memory Usage.

- Open terminal in kali linux.
- Type the command **free -h**.
- Output is as follows.

```
(kali㉿kali)-[/home/kali]
PS> free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	929Mi	177Mi	28Mi	1.0Gi	1.0Gi
Swap:	953Mi	141Mi	811Mi			

Step #03: Command free -h

This will display current memory RAM.

• STEP # 04 •

Step # 04 : Run command for Manage a Service.

- Open terminal in kali linux.
- Type the command **sudo systemctl start/stop/status apache2**.
- Output is as follows.

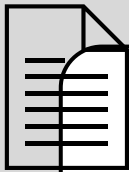
```
(kali㉿kali)-[~]
$ sudo systemctl start apache2

(kali㉿kali)-[~]
$ sudo systemctl stop apache2

(kali㉿kali)-[~]
$ sudo systemctl status apache2
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: https://httpd.apache.org/docs/2.4/

Jul 13 04:09:04 kali apachectl[56576]: AH00558: apache2: Could not reliably
Jul 13 04:09:04 kali systemd[1]: apache2.service: Deactivated successfully.
Jul 13 04:09:04 kali systemd[1]: Stopped apache2.service - The Apache HTTP S
Jul 13 04:09:12 kali systemd[1]: Starting apache2.service - The Apache HTTP
Jul 13 04:09:12 kali apachectl[56664]: AH00558: apache2: Could not reliably
Jul 13 04:09:12 kali systemd[1]: Started apache2.service - The Apache HTTP S
Jul 13 04:09:19 kali systemd[1]: Stopping apache2.service - The Apache HTTP
Jul 13 04:09:19 kali apachectl[56741]: AH00558: apache2: Could not reliably
Jul 13 04:09:19 kali systemd[1]: apache2.service: Deactivated successfully.
Jul 13 04:09:19 kali systemd[1]: Stopped apache2.service - The Apache HTTP S
lines 1-15/15 (END) ... skipping ...
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: https://httpd.apache.org/docs/2.4/
```

Step #04: Command sudo systemctl start/stop/status apache2



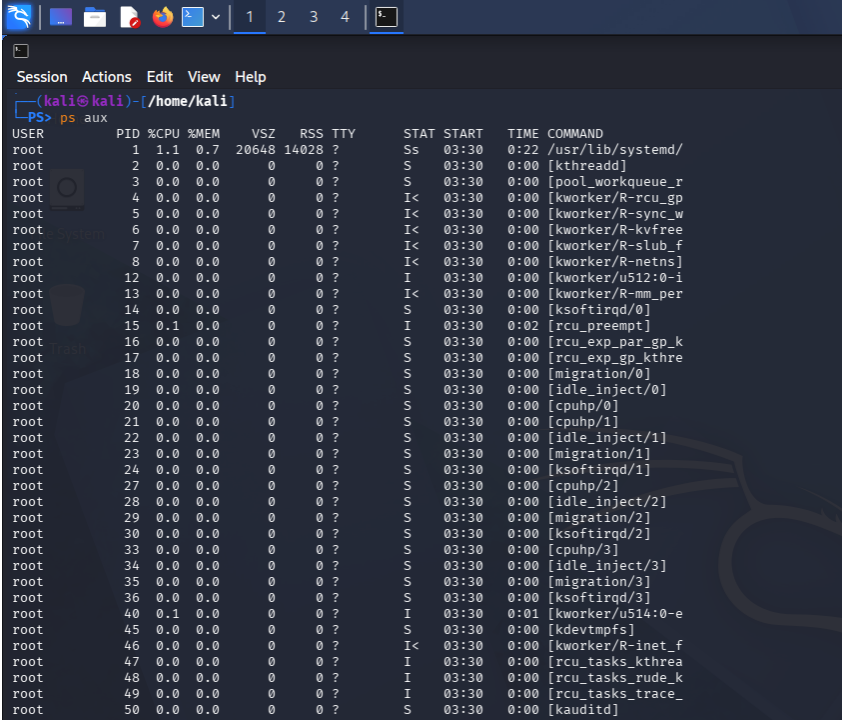
This will perform the operation depending on the command.



• STEP # 05 •

Step # 05 : Run command for Monitor Running Processes.

- Open terminal in kali linux.
- Type the command **ps aux**.
- Output is as follows.



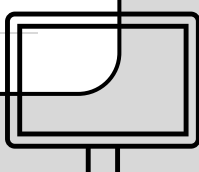
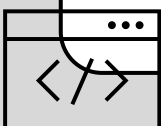
```
(kali@kali)-[/home/kali]
PS> ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root            1  1.1  0.7 20648 14028 ?        Ss   03:30   0:22 /usr/lib/systemd/
root            2  0.0  0.0      0     0 ?        S    03:30   0:00 [kthreadd]
root            3  0.0  0.0      0     0 ?        S    03:30   0:00 [pool_workqueue_r
root            4  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-rcu_gp
root            5  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-sync_w
root            6  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-kvfree
root            7  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-slub_f
root            8  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-netns]
root           12  0.0  0.0      0     0 ?        I    03:30   0:00 [kworker/u512:0-i
root           13  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-mm_per
root           14  0.0  0.0      0     0 ?        S    03:30   0:00 [ksoftirqd/0]
root           15  0.1  0.0      0     0 ?        I    03:30   0:02 [rcu_preempt]
root           16  0.0  0.0      0     0 ?        S    03:30   0:00 [rcu_exp_par_gp_k
root           17  0.0  0.0      0     0 ?        S    03:30   0:00 [rcu_exp_gp_kthre
root           18  0.0  0.0      0     0 ?        S    03:30   0:00 [migration/0]
root           19  0.0  0.0      0     0 ?        S    03:30   0:00 [idle_inject/0]
root           20  0.0  0.0      0     0 ?        S    03:30   0:00 [cpuhp/0]
root           21  0.0  0.0      0     0 ?        S    03:30   0:00 [cpuhp/1]
root           22  0.0  0.0      0     0 ?        S    03:30   0:00 [idle_inject/1]
root           23  0.0  0.0      0     0 ?        S    03:30   0:00 [migration/1]
root           24  0.0  0.0      0     0 ?        S    03:30   0:00 [ksoftirqd/1]
root           27  0.0  0.0      0     0 ?        S    03:30   0:00 [cpuhp/2]
root           28  0.0  0.0      0     0 ?        S    03:30   0:00 [idle_inject/2]
root           29  0.0  0.0      0     0 ?        S    03:30   0:00 [migration/2]
root           30  0.0  0.0      0     0 ?        S    03:30   0:00 [ksoftirqd/2]
root           33  0.0  0.0      0     0 ?        S    03:30   0:00 [cpuhp/3]
root           34  0.0  0.0      0     0 ?        S    03:30   0:00 [idle_inject/3]
root           35  0.0  0.0      0     0 ?        S    03:30   0:00 [migration/3]
root           36  0.0  0.0      0     0 ?        S    03:30   0:00 [ksoftirqd/3]
root           40  0.1  0.0      0     0 ?        I    03:30   0:01 [kworker/u514:0-e
root           45  0.0  0.0      0     0 ?        S    03:30   0:00 [kdevtmpfs]
root           46  0.0  0.0      0     0 ?        I<   03:30   0:00 [kworker/R-inet_f
root           47  0.0  0.0      0     0 ?        I    03:30   0:00 [rcu_tasks_kthrea
root           48  0.0  0.0      0     0 ?        I    03:30   0:00 [rcu_tasks_rude_k
root           49  0.0  0.0      0     0 ?        I    03:30   0:00 [rcu_tasks_trace_
root           50  0.0  0.0      0     0 ?        S    03:30   0:00 [kauditd]
```

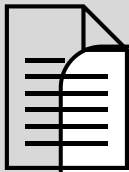
Step #05: Command ps aux

This will display all running processes with their info.

Assignment Conclusion:

In this assignment we ran some commands in the kali linux terminal to get a better understanding of managing an OS from the terminal. This increased and polished my ability to use and learn more about cybersecurity.





• The End •

